

Trustworthy-PWS: An Operator-Agnostic Public Warning System (PWS) Security Architecture in 5G

Sourav Purification, Sang-Yoon Chang
University of Colorado Colorado Springs, {spurific, schang2}@uccs.edu

Abstract—Public Warning System (PWS), a critical architecture for public safety and emergency alerting in cellular networking, faces a fundamental *PWS Trilemma*: it must simultaneously ensure universal availability, real-time efficiency, and resilient trust. Current cellular PWS implementations and prior research fail to achieve a balance between these conflicting objectives. Existing research in cellular networking security relies on the mobile network operator to build trust and therefore only works within subscribed network (violating the PWS design goal of universal availability) and becomes ineffective in the presence of the operator’s base station’s failures (whether due to malicious compromise or simply the absence of infrastructure within a given geographic area). To resolve the PWS Trilemma, this study proposes *Trustworthy-PWS*, a novel *operator-agnostic* security architecture that decouples alert verification from the mobile network operator. By establishing the public authority (alert initiator) as the independent root of trust and leveraging opportunistic key caching via a secure NAS layer, our architecture enables all users, including roaming and unauthenticated users, to cryptographically verify alerts directly from the public authority, bypassing base stations. To address the strict latency constraints of emergency alerts, we introduce a lightweight commitment-based protocol using a reverse hash chain that reduces the verification complexity from linear $\mathcal{O}(n)$ to constant $\mathcal{O}(1)$. We rigorously validate the architecture using the Tamarin Prover and formally verify the source authentication and resilience against compromised infrastructure. A prototype implementation on a Software-Defined Radio (SDR) testbed demonstrates an 85% reduction in cryptographic latency compared to standard signatures, proving that the system ensure universal availability, security and real-time performance thus resolves PWS Trilemma.

Index Terms—Public Warning System, PWS, 5G Security, Broadcast Authentication, Cell Broadcast Service, Formal Verification

I. INTRODUCTION

The Public Warning System (PWS) is a 3GPP-standardized [1] cellular broadcast service designed to broadcast critical alerts for national emergencies (e.g., presidential alerts), natural disasters (e.g., earthquakes and tsunamis), and public safety threats (e.g., AMBER alerts). PWS alerting is a sporadic event, but is designed to disseminate time-sensitive warnings to millions of users instantly upon its trigger. For example, earthquake and tsunami alerts must reach to users within 4 seconds of alert trigger [2]. Unlike commercial cellular services (e.g., voice and data), which are delivered exclusively to subscribed users, PWS is operator- and subscription-agnostic so that a user equipment can receive PWS from any mobile network operators and without paid subscriptions [3]. The design

ensures the *universal availability* of alerts to every mobile user, regardless of the availability of the subscribed network. The universal availability supports the critical applications of PWS, including disaster and emergency communications.

In PWS, a public authority initiates an alert and relies on unauthenticated broadcast communication from the base stations of operator networks to users for efficient alert delivery. Although the data and control traffic of subscription-based cellular services are secured via mutual authentication and key agreement between the user device and its subscribed network [4], PWS alerts are transmitted using open radio broadcasting without source authenticity or message integrity [5]. Such design provides two benefits: First, it ensures universal availability, allowing users to receive alerts from any available network without the latency of cryptographic handshakes. Second, it ensures timely delivery because omitting heavy verification operations reduces the processing overhead during time-sensitive emergencies [2]. However, this lack of authentication exposes the system to severe security risks [6], [7]. An adversary can exploit this unauthenticated broadcasting by deploying a fake base station or compromising legitimate base stations to broadcast fabricated alerts, potentially inciting a mass panic and actions.

Prior research to address such authentication deficiency has relied on the mobile network operator to serve as the source of trust, utilizing its public keys. However, such approach results in two critical limitations. First, such approaches becomes ineffective in the presence of the operator’s base station’s failures, whether due to malicious compromise or simply the absence of infrastructure within a given geographic area. Second, state-of-the-art solutions violate the PWS design goal of universal availability, since it supports PWS authentication only within subscribed networks. They do not consider universal availability but rather only focus on PWS integrity and its trustworthy delivery. Since the previous research solutions require to be within subscribed network, their PWS delivery is more limiting than the current PWS practice designed for and prioritizing universal availability. More specifically, while the current PWS implementation delivers the PWS alerts across all mobile network operators as well as during roaming, limited service state (LSS), and to non-3GPP users, prior research does not work in such scenarios.

Prior research therefore fails to address the *PWS Trilemma* presenting a conflict between universal availability, system efficiency (latency and bandwidth), and a resilient trust model. Standard Public Key Infrastructure (PKI)-based solutions [8],

[9], [10] or 3GPP standardization efforts [11] typically anchor trust in the preinstalled network credentials in the subscription module chip within the user device. This approach fundamentally violates universal availability because it cannot authenticate alerts from a foreign operator owing to the lack of anchor trust. In contrast, lightweight protocols, such as TESLA [12], [13] designed for broadcasting in lossy channels, fail to meet the system efficiency requirements for emergency alerts. Their reliance on delayed key disclosure introduces unacceptable latency for time-sensitive warnings (e.g., earthquakes), whereas their dependence on time synchronization renders them vulnerable to manipulation by compromised base stations. Furthermore, previous research assumes base stations are trusted entities. Being geographically distributed and physically exposed, base stations are considered as susceptible to compromise [14]. Relying trust anchor in these base stations makes the solution vulnerable to compromised infrastructure. Our approach anchors the trust in the PWS public authority (the PWS initiator, separate from the mobile network operators), utilizing its public key to design and build efficient authentication verification. Our approach enables delivery of the PWS alerts without relying on a singular mobile network operator while protecting security so that only the authenticated PWS is accepted and further processed, e.g., denying false PWS alert transmissions by fake base station.

To address the shortcomings of existing research, our study proposes *Trustworthy-PWS (T-PWS)*, an operator-agnostic PWS security architecture. We transform the PWS Trilemma and threat vectors into concrete design principles to build a trustworthy alerting system. Our goal is to guarantee the alert source authenticity for all users while minimizing delivery latency. The specific contributions of this study are as follows:

- **Operator-Agnostic Trust Architecture:** We propose T-PWS, a novel operator-agnostic trust model that allows users to verify alerts from any network provider, decoupling security from the mobile network. The trust model uses the public authority as the root of trust, ensuring universal availability. By opportunistically caching the authority's public key, user devices can verify alerts received from any mobile network operator.
- **Lightweight Protocol Design:** The T-PWS security protocol uses one digital signature and multiple hash operations that reduces verification complexity to $\mathcal{O}(1)$ from $\mathcal{O}(n)$ for n number of alert segments. This lightweight protocol design satisfies strict latency requirements and fits within the limited bandwidth of 5G radio broadcasts.
- **Comprehensive Validation:** We rigorously validate the security and the performance of the T-PWS using dual-layer validation. First, we prove the security of the protocol using Tamarin-Prover [15] to prove the security functions of our scheme in an operator-agnostic setting. Second, we validate the lightweight aspect of the architecture through a prototype implementation on a software-defined radio (SDR) testbed. The results show that T-PWS reduces end-to-end computation latency and energy

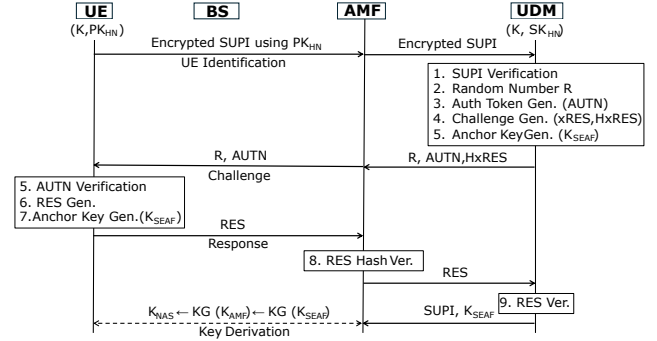


Fig. 1: 5G-AKA protocol in an abstracted form. It shows the identification, challenge-response, and NAS layer key derivation phases. We omit mathematical functions behind each operation for simplicity. The arrow labels show only the payloads for brevity and clarity.

consumption in verifying alerts compared to standard baseline. The wireless experiment shows that the communication overhead due to cryptographic algorithms has a negligible impact on alert receiving and processing alerts at users compared to the channel noise.

II. BACKGROUND

In this section, we describe the background of the 5G cellular network and PWS.

A. 5G Network: Architecture, Trust and Authentication

1) *5G Network Architecture and Protocol Stack:* The 5G architecture comprises User Equipment (UE), Base Station (BS), and Core Network (CN). Within the core network, we distinguish between the *Home Network (HN)*, which holds the subscriber's permanent credentials (in the UDM), and the *Serving Network (SN)*, which manages mobility via the Access and Mobility Management Function (AMF). The protocol stack operates on two distinct layers: the *Radio Resource Control (RRC)* layer manages the physical radio channel and broadcasting (including PWS) between the UE and BS. In contrast, the *Non-Access Stratum (NAS)* layer handles the secure control signaling between the UE and AMF.

2) *5G Authentication and Key Agreement Protocol Between CN and UE:* The roots of trust in 5G is provided by the shared keys between UE and UDM/HN, which are a long-time symmetric key K and HN's public key PK_{HN} . These keys are delivered to the UE via Universal Subscriber Identity Module (USIM), a physical hardware chip installed at the UE before the UE ever connects to any 5G network. Figure 1 shows these keys (K, PK_{HN}) shared between UE and UDM/HN on the top of the figure. It also shows the rest of the 5G Authentication and Key Agreement (AKA) protocol handshake dependent on K and PK_{HN} . 5G AKA establishes a hierarchical cryptographic keys, specifically the NAS encryption and integrity key (K_{NAS}). While NAS messages are physically encapsulated within RRC frames for transmission over the radio interface, this key secures the NAS connection between the UE and

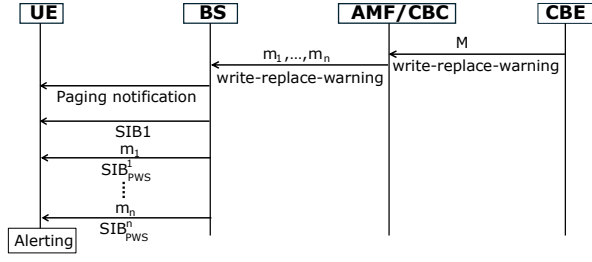


Fig. 2: PWS protocol message flow from alert initiator (CBE on the rightmost) to alert receiver (UE on the leftmost) via mobile network (two middle nodes). The alert M is segmented into n CB pages, which are mapped to n instances of SIB_{PWS} .

the AMF. Our scheme, T-PWS leverages this pre-established secure channel to provision the T-PWS trust anchor (the CBE's public key) to the UE.

3) *User Service States*: Based on 5G-AKA authentication status, UEs are categorized into two primary service states: normal and limited. The Normal Service State is applied to UEs that have successfully established a security key (K_{NAS}) with the AMF. This includes *Subscribed Users* (connected to the HN), *Roaming Users* (connected to a visited SN), and *Non-3GPP Users* (e.g. setup NAS via Wi-Fi). In contrast, the *Limited Service State (LSS)* is applied when a UE cannot complete authentication due to the lack its subscribed network coverage. LSS devices cannot access voice or data cellular services; however, they can camp on any available mobile network base station to receive emergency PWS broadcasts.

B. Public Warning System (PWS)

The PWS facilitates the emergency alerts delivery from public authority to UEs within a target geographic area via mobile network. we describe the PWS architecture and alert flow of the 5G PWS in the following sections.

1) *Architecture*: The 3GPP-standardized PWS architecture comprises three main stakeholders: the Public Authority, the Mobile Network, and the UE. The alert flow begins with the Public Authority, which identifies the emergency and initiates the alert via a functional node called the Cell Broadcast Entity (CBE). The Mobile Network bridges the connection between the authority and users within the targeted geographic area. The Cell Broadcast Center (CBC) of the serving network handles the broadcasting functionality within the mobile network and provides an interface with the CBE and AMF. The AMF and base stations of the mobile network are involved in identifying the target alerting area and broadcasting alerts.

2) *PWS Alert Segmentation, Formatting and Scheduling*: The alert message, M (usually 82 to 1230 bytes), initiated by the CBE is segmented, formatted, and scheduled by intermediary network entities. More specifically, the alert content is not delivered transparently as an intact payload to the UEs; rather, the intermediary network entities segment the alerts into fixed sizes, format the alerts into new message types, and broadcast transmissions on a schedule. The CBE identifies

the alert type, generates the alert content, defines the target geographical scope, and forwards the request to the CBC using a `write-replace-warning` message.

The CBC formats the alert by segmenting into 1 to 15 Cell Broadcast (CB) pages of 82 bytes each. Each CB page is identified by a segment number, and each alert is identified by a serial number for complete alert reconstruction upon reception at UE. As shown in Figure 2, CBC segments an alert M , received from CBE, into n CB pages ($m_1, \dots, m_n$). The CBC then formats each CB page into a `write-replace-warning` message and delivers it to the AMF. The AMF identifies the alerting area and forwards the CB pages to the specific base stations of the target alerting area.

The base station translates the CB pages into byte-size limited (372 bytes) system information block type x (SIB_x) messages depending on the type of alert. For example, in 5G, SIB_6 carries the primary earthquake and tsunami notifications, and SIB_7 carries the secondary notifications. In this paper, we denote PWS-specific SIBs as SIB_{PWS} without differentiating between alert types. The base station maps the segmented CB pages into multiple SIB instances of the same SIB, separated by the segment number. Each of these CB pages is mapped to an arbitrary type of SIB, SIB_{PWS}^k , where $1 \leq k < n$. The base station does not broadcast SIB_{PWS}^k directly, rather schedules the intention of alert broadcast within the SIB_1 . After formatting, the base station employs broadcasting communication at the RRC layer to distribute the alert schedule in SIB_1 and alert contents in SIB_{PWS}^k .

3) *PWS Alert Broadcasting*: Alerts are scheduled in SIB_1 and follow a strict scheduling sequence, as illustrated in Figure 2. To receive alerts, the UE must camp on any available cell of a base station and periodically scan SIB_1 at the RRC layer. SIB_1 is the essential broadcast information message that the base station transmits with a 160 ms periodicity that contains the scheduling information (periodicity and time-domain window) for the SIB_{PWS} . Upon reception of SIB_1 , the UE determines the periodicity and time-domain reception window of the upcoming SIB_{PWS} . It follows the designated time window to receive an actual alert content in SIB_{PWS} . For multi-segment alerts received in SIB_{PWS}^k , the UE continues to monitor the PWS scheduling window to receive all subsequent segments and reconstructs the complete warning message using the segment number.

III. PROBLEM STATEMENT AND RELATED WORK

Emergency alerts are universally available to everyone within the target area and are critical for timely delivery of information. This requires the PWS infrastructure to rely on a broadcasting mechanism to disseminate alerts to the mass population. While alerts are delivered rapidly to a large number of people, the current PWS system remains vulnerable to spoofing, injection, and false alerts [7], [6].

Designing a secure PWS is uniquely challenging because the system must satisfy three conflicting objectives: security, availability, and efficiency, creating a *PWS Trilemma*. To secure the PWS architecture, the solution must provide

authenticity and integrity of the alert broadcast against a strong adversary, operate under strict bandwidth and latency requirements, and most importantly, the alert must be available to all UEs in the target area. In this section, we analyze the vulnerability landscape in PWS, discuss the intricate challenges, and critically evaluate the state-of-the-art solution against the PWS Trilemma.

A. Vulnerabilities in PWS and RRC Broadcasting

The lack of authentication and integrity protection in RRC broadcasting messages, including SIBs and paging messages, are the prime reasons for PWS insecurity. Hussain et al. exposed vulnerabilities in paging messages [16] that an attacker can exploit for fake alert injection [17], [18]. Lee et al. [7] demonstrated the first practical spoofing attacks on PWS alerts. Their study investigated the vulnerability of PWS broadcasting to dispatch fake presidential alerts using software-defined radios and the available open-source software in 4G-LTE. They demonstrated that UEs blindly accept alerts from any alerting base station without validating the source authenticity. Bitsikas et al. [6] conducted a comprehensive analysis of PWS security in 5G cellular networks. Despite 5G providing enhanced security, their work identified several security flaws in PWS that an attacker can exploit to send fake PWS alerts, including insecure paging and system information messages. The attacker exploits these vulnerabilities via a malicious attachment [19] or by setting up a fake base station that broadcasts crafted SIBs with higher signal power, forcing the victim UEs to decode and display fabricated warnings [20], [21].

B. PWS Trilemma

We define the *PWS Trilemma* as a set of three conflicting objectives that a functional PWS architecture must simultaneously satisfy. Current solutions typically sacrifice one to achieve the others.

1) *Universal Availability*: As a regulatory public safety service, PWS must be available to all users within the target geographic area [22], [23], regardless of subscription status. This includes UEs in the normal service state and crucially in the limited service state, those without a valid USIM or network coverage. Because limited-service state UEs lack the cryptographic credentials to authenticate with a foreign network, any security solution that anchors trust in the USIM (e.g., standard 5G-AKA) inherently excludes this vulnerable population from verifying alerts from a foreign network, violating the primary functional requirement of universal availability.

2) *Real-Time Efficiency*: The PWS operates under strict latency and bandwidth bounds. Functionally, alerts such as earthquake warnings must be delivered within 4 seconds of generation [2]. Cryptographically, this precludes heavy verification operations that can introduce processing delays. The physical layer of the 3GPP standard limits the payload size of SIB broadcasting, making it infeasible to append standard digital signatures to every broadcast segment without incurring prohibitive fragmentation and transmission overhead.

3) *Resilient Trust*: A resilient PWS must guarantee end-to-end source authentication and message integrity irrespective of the alert delivery network. However, most existing security solutions anchor trust within the mobile network, assuming that it operates as a fully trusted infrastructure where adversaries are restricted to the air interface (e.g., fake base stations). This network-centric trust model violates system resilience in two ways. First, it binds alert verification exclusively to the user's subscribed network, directly conflicting with the mandate of universal availability. In the event of an emergency, a device cannot seamlessly verify alerts broadcast by foreign or unauthenticated networks. Second, blindly trusting the delivery infrastructure is inherently fragile; if an adversary compromises a physically exposed base station, they can leverage network-level trust to inject authenticated but fabricated alerts.

C. Related Research and Limitations

Recent research has proposed different cryptographic approaches for securing broadcast SIB messages. In this section, we categorize them and analyze their limitations against the PWS Trilemma described in Section III-B.

1) Public Key and Digital Signature-Based Approaches:

Recent research has focused on securing SIB broadcasting messages by establishing a base station public key for digital signature-based message authentication [8], [24], [9], [10]. Hussain et al. [8] and Ross et al. [9] proposed a PKI-based solution that delivers the base station public key to UEs and leverages the HN public key as the root of trust installed in the USIM card. Singla et al. [24] adopted a similar approach but used identity-based cryptography. Their approach adds significant data bytes to broadcasting because it includes both the public key and digital signature within the broadcasting messages. In contrast, Purification et al. [10] adopted a blockchain-based approach in base station public key delivery that requires transmitting only signatures in broadcasting, although the solution relies on pre-installed core network credentials in the USIM card. Purification et al. [25] proposed a computationally efficient way to enable PWS alert verification using a commit-and-reveal technique that omits digital signature-based verification of broadcast messages during alerting.

Although these solutions address RRC broadcasting vulnerabilities in general, they do not consider the trilemma. Relying on the root key installed in the USIM card limits the UEs to authenticate only its subscribed and roaming networks, leaving limited service state and non-3gpp users unable to authenticate the messages, thus violating universal availability. Their threat model considers the base station to be a trusted entity that signs the SIB broadcasting.

2) *Lightweight Broadcast Authentication*: Timed Efficient Stream Loss-tolerant Authentication (TESLA) [26], [13], [12] is a widely recognized lightweight broadcast authentication protocol specifically designed for lossy channels, such as those in wireless networks. The TESLA protocol employs a one-way key chain [27], [28] based on a random number to generate symmetric keys, which are then revealed in a delayed time

in reverse order for the authentication of broadcast messages. The bootstrapping of TESLA requires sender authentication using a public-key-based digital signature and sharing of a key disclosure schedule. Following this authentication, the sender reveals the one-time key in reverse order at an agreed time interval. Inherently, TESLA fails to satisfy the efficiency constraint owing to the mandatory delay for key disclosure, which induces latency in time-sensitive alerts, such as those for natural disasters. Furthermore, TESLA requires a public key or root key for bootstrapping, it cannot guarantee universal availability to limited service state and non-3gpp users because of the trust model constraint.

3) *Standardization Effort*: 3GPP standardization studies SIB broadcast security in two key studies. To ensure security against fake base stations, TR 33.809 [11] proposes signing SIB messages using a digital signing function implemented in the core network. In PWS security, TR 33.855 [29] discusses the end-to-end security of alert messages against a compromised base station. One candidate solution involves provisioning the CBE public key via the NAS layer and validating it using a root key installed in the USIM card. While these reports correctly identify the threat of compromised infrastructure, the proposed solutions remain theoretical studies owing to unresolved conflicts with the PWS Trilemma.

IV. THREAT MODEL AND ASSUMPTIONS

In this section, we first describe the adversary model with security assumptions, followed by defining the solution requirements for PWS Security that guide our design.

A. Threat Model

1) *Adversary Model*: We model the adversary as a Dolev-Yao [30] attacker with complete control over the radio interface between the UE and base station. The adversary can intercept, modify, inject, and replay messages on the air radio interface. We also extend this model to include compromised base stations, which allows the adversary to retrieve the base station keys. This capability enables an adversary to exploit the insecure broadcasting of the PWS to inject fake alerts. In its sophisticated form, the adversary can perform a *wormhole attack*, capturing a valid alert broadcast in a location and broadcast the alert to another location. The wormhole attack is relevant to PWS because the alerts are geographically bounded using the `Warning Area Coordinates` field in the alert, allowing the adversary to change the target area information and modify the scope of the alerting area.

2) *Security Assumptions*: We assume that the public authority (CBE) and Core Network infrastructure (AMF, CBC and UDM) are trusted domain. Inter-entity communication, particularly the critical CBE-CBC link, is assumed to be protected using standard protocols (e.g., IPsec [31]). Consequently, we exclude insider threats within the core network, focusing on the adversary's capabilities strictly on the wireless interface and untrusted base stations.

We assume that standard cryptographic primitives (e.g., hash functions and digital signatures) are secure against the threat

model. Furthermore, we rely on the 5G-AKA protocol to establish a secure NAS channel for provisioning the CBE public-key. While we acknowledge the known vulnerabilities of 5G-AKA regarding user privacy and DoS [32], our architecture depends solely on the protocol's provision of integrity and authentication for the NAS layer connection, which remains robust against these specific threats.

B. Solution Requirements

To resolve the PWS Trilemma described in Section III-B and threats described in Section IV-A, we derive the following specific availability, trust, and efficiency requirements:

1) *Requirements for Universal Availability*: To satisfy the first leg of the PWS Trilemma, the solution must function independently of the user's subscription status.

- **F1. Operator-Agnostic Verification**: The solution must enable alert verification for all device states, including Subscribed, Roaming, Limited Service State, and Non-3GPP access.
- **F2. Decoupled Trust Model**: To support F1, the verification process must not rely on the Home Network credentials (USIM) or active network authentication (e.g., 5G-AKA). Verification must be self-contained, using a trust anchor independent of the distribution network.

2) *Requirements for Resilient Trust*: To satisfy the second leg, the solution must ensure end-to-end alert authentication under the adversary defined in Section IV-A.

- **S1. Secure Trust Bootstrapping**: The system must securely provision and update the CBE's public key (Trust Anchor) to the UE to ensure that the UE can trust the CBE. This process must be resilient against man-in-the-middle (MitM) attacks and prevent the injection of malicious keys by compromised base stations.
- **S2. End-to-End Source Authentication**: The UE must be able to cryptographically verify that the alert is originated from the legitimate CBE and was not modified by the adversary.
- **S3. Geo-Fencing**: The signature must cryptographically bind the alert to its target, the *Warning Area* via Geo-Fencing. The UE must be able to detect and reject *Wormhole* attacks, where valid alerts are broadcast to incorrect geographical locations.
- **S4. Replay Resistance**: The solution must ensure message freshness to prevent the rebroadcast of valid but expired alerts.

3) *Requirements for System Efficiency*: To satisfy the third leg, the security overhead must not violate the strict operational constraints of the 5G PWS.

- **P1. Latency Compliance**: The cryptographic overhead must not jeopardize the regulatory requirement for time-sensitive alert delivery. Verification must be near-instantaneous upon receipt.
- **P2. Standard & Bandwidth Compliance**: The security metadata must fit within the strict payload limits of existing System Information Blocks (SIBs) (e.g., < 372

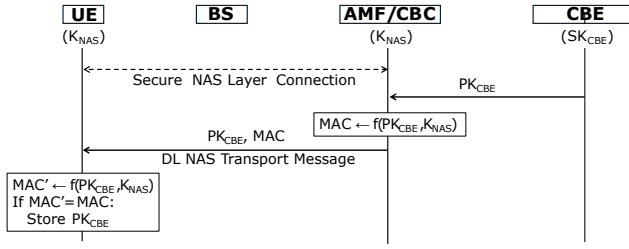


Fig. 3: Protocol message flow of T-PWS Trust Establishment in Pre-Alert. The CBE public key is delivered to UEs after NAS security setup during re/registration process.

bytes for SIBs) to ensure backward compatibility and minimal radio resource usage.

- **P3. Energy Efficiency:** The verification algorithm must be lightweight to minimize battery consumption on resource-constrained UEs for alert verification.

V. OUR TRUSTWORTHY-PWS (T-PWS) SCHEME

We propose *Trustworthy-PWS (T-PWS)*, an operator-agnostic secure public warning architecture that decouples the root-of-trust from the mobile network. The protocol operates in two distinct asynchronous phases: Trust Establishment in pre-alert and PWS Delivery and Authentication during alert. In the Trust Establishment phase, the CBE public key is securely provisioned to UEs via the NAS layer at any time before alerting. During alerting, the Secure Alert Broadcast phase employs bandwidth-efficient authentication using reverse hash chains and digital signatures.

A. T-PWS Trust Establishment in Pre-Alert

Because UE verifies the PWS signed by CBE, T-PWS involves the public and private keys of CBE, which are PK_{CBE} and SK_{CBE} , respectively. In pre-alert, T-PWS leverages the existing 5G AKA protocol and the resulting channel to securely deliver and authenticate PK_{CBE} . As described in Section II-A2, 5G AKA results in K_{NAS} , which symmetric key is then used to build an encryption-based Secure NAS Layer Connection for confidentiality protection. T-PWS in pre-alert delivers PK_{CBE} as payloads of the DL NAS connection using the Secure NAS Layer Connection channel, as described in Figure 3. The rest of this section describes in greater details further scalability and key management benefits of using the CBE’s public key to anchor T-PWS (Section V-A1) as well as how T-PWS operates in pre-alert to enable the authenticated T-PWS deliveries in subscribed, roaming, and non-3GPP access (Section V-A2) and in Limited Service State or LSS via opportunistic caching (Section V-A3).

1) *Scalability and Key Management of T-PWS:* To ensure scalable provisioning, revocation, and cross-jurisdiction operation, T-PWS models the CBE as a centralized alerting gateway (similar to FEMA’s IPAWS in the US [33]). The architecture assumes a single CBE as national trust anchor per country. UE only needs to provision and store a single public key

(PK_{CBE}) for its current national jurisdiction. Domestic cross-jurisdiction mobility (e.g., moving between states) requires no cryptographic updates because alerts are delivered via single CBE entity. During international roaming, the AMF of the visited network simply provisions the visited country’s specific PK_{CBE} to UE upon initial NAS attachment. Furthermore, because the singular root key (SK_{CBE}) is managed by a federal entity within a secure facility, the risk of key compromise is centralized and minimized. In the event of revocation, operators need only push a single updated key via standard NAS configuration updates.

2) T-PWS for subscribed, roaming and non-3gpp users:

As described in Section II-A3, users in the normal service state (subscribed, roaming, and non-3GPP access) can participate in the 5G-AKA protocol and thus establish a Secure NAS Layer Connection. Because this secure layer is established between the AMF and the UE irrespective of the underlying radio infrastructure (e.g., base station or Wi-Fi), it guarantees that normal-state users are securely provisioned with PK_{CBE} upon initial network attachment. As illustrated in Figure 3, T-PWS delivers the CBE public key, PK_{CBE} , using an integrity-protected DL-NAS Transport message as follows:

$$AMF \rightarrow UE : \{PK_{CBE}\}_{K_{NAS}}$$

UEs then store PK_{CBE} within secure persistent storage of the device for alert verification during an emergency.

3) *T-PWS for Edge-case UEs (LSS UEs):* For edge cases, such as UEs in a limited service state (LSS), T-PWS employs an *Opportunistic Caching* strategy, similar to *Trust on First Use (TOFU)* [34]. LSS is typically a temporary situation for users with expired subscriptions or loss of network coverage, as explained in Section II-A3. We assume that UEs will naturally be in a normal service state (as a subscribed, roaming, or non-3GPP access user) and will securely obtain and persistently store PK_{CBE} prior to entering the LSS state. If the UE subsequently loses service, it retains the cached PK_{CBE} to verify emergency alerts. This effectively extends the trust established during the last successful network connection to cover the disconnection period. We exclude “LSS cold start” scenarios, where uninitialized devices lack PK_{CBE} , from our scheme, as they represent a statistically negligible edge case compared to typical device lifecycles.

B. PWS Delivery and Authentication During Alert

In an emergency, we employ a lightweight commitment-based protocol to authenticate the broadcast. To circumvent the bandwidth constraints of SIBs and avoid the latency of signing every segment, we employ a reverse hash chain that leverages the lightweight efficiency of hashing for message integrity and a digital signature for alert source authenticity. T-PWS introduces computation at CBE and at UE, and security parameters within the standard protocol messages as illustrated in Figure 4.

1) Hash Chain and Digital Signature Construction at CBE:

Let M be the alert message at the CBE dispatch. The CBE

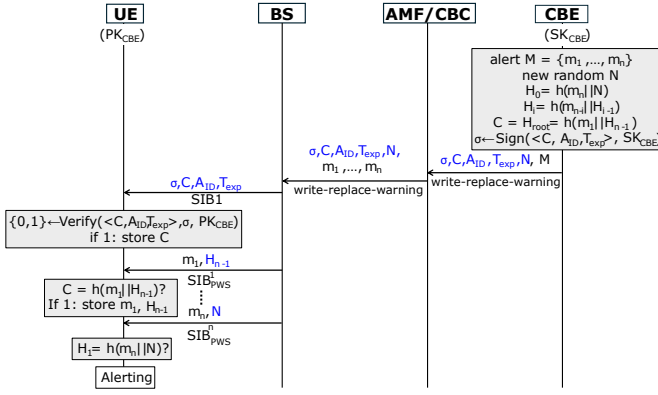


Fig. 4: Protocol message flow of Secure Alert Broadcast phase (during alert) of T-PWS build on 3GPP-Standard (shown in Figure 2). T-PWS specific computations are shown within gray rectangle while protocol data is shown in blue color.

logically segments M into n pages: $M = \{m_1, \dots, m_n\}$. It also generates a fresh nonce N and computes a reverse hash chain as follows:

$$\begin{aligned} H_0 &= h(m_n || N) \\ H_i &= h(m_{n-i} || H_{i-1}) \quad \text{for } 1 \leq i < n \\ C &= H_{\text{root}} = h(m_1 || H_{n-1}) \end{aligned}$$

The root hash, H_{root} , serves as the cryptographic commitment, C . To bind this commitment to the context, the CBE generates a digital signature σ using its private key SK_{CBE} over the commitment C , the alerting area ID (A_{ID}), and the alert expiration time (T_{exp}):

$$\sigma \leftarrow \text{Sign}(\langle C, A_{\text{ID}}, T_{\text{exp}} \rangle, SK_{\text{CBE}})$$

The alerting area ID, A_{ID} , is the target geographic area where the alert needs to be delivered. A_{ID} can be the Warning Area Coordinates information field included in the warning message and acts as a geo-fencing for UE to determine if the UE is located in the alerting area. T_{exp} is the validity time of the alert determined by the CBE. A_{ID} and T_{exp} inside the digital signature ensure that these values are authenticated and used for wormhole resistance and replay defense, respectively.

After the hash chain and digital signature generation, the CBE sends σ , C , A_{ID} , T_{exp} , N , M and other parameters to the CBC. CBC follows the standard alert segmentation procedure described in Section II-B2. It forwards all these CB pages along with the security parameters to the target AMF. The AMF then forwards the CB pages encoded in the Write-Replace-Warning message to the base stations.

2) *Alert Broadcast Scheduling by Base Station*: The base station receives n CB pages $m_1, \dots, m_n$, σ , A_{ID} , T_{exp} , and N . The BS reconstructs the hash chain ($H_1, \dots, H_n$) similar to CBE and parses the CB pages to map them to PWS-specific SIBs, $\text{SIB}_{\text{PWS}}^k$, as described in Section II-B3. It cannot modify the content m_i or the chain links H_i without invalidating the

root commitment C , which is signed by the CBE. The base station broadcasts $\{C, \sigma, A_{\text{ID}}, T_{\text{exp}}\}$, the security header of the alert message, in the essential SIB1 broadcast message, followed by the sequential $\text{SIB}_{\text{PWS}}^k$ according to the scheduling information provided in SIB1. Each $\text{SIB}_{\text{PWS}}^k$ payload contains the message segment m_k and the *previous* hash link H_{n-k} , enabling the immediate verification of the *next* segment. In our scheme, only SIB1 includes a signature (σ), whereas each $\text{SIB}_{\text{PWS}}^k$ includes a lightweight hash link, except for the final $\text{SIB}_{\text{PWS}}^n$ contains N . The base station acts as a transparent relay and broadcasts SIBs, as illustrated in Fig. 4. $\text{SIB}_{\text{PWS}}^{(k)}$ transmits n number of CB pages according to the transmission schedule mentioned in the SchedulingInfo of SIB1.

$$\text{BS} \xrightarrow{\text{SIB1}} \forall \text{UE} : \{C, \sigma, A_{\text{ID}}, T_{\text{exp}}\}$$

$$\text{BS} \xrightarrow{\text{SIB}_{\text{PWS}}^k} \forall \text{UE} : \{m_k, H_{n-k}\} \quad \text{for } 1 \leq k < n$$

3) *Alert Verification at UE*: The UE begins the alert verification process immediately after receiving SIB1 along with the alert message security header, $\{C, \sigma, A_{\text{ID}}, T_{\text{exp}}\}$. The UE verifies the security header using PK_{CBE} . Successful security header verification follows the waiting to receive and verify $\text{SIB}_{\text{PWS}}^k$.

During security header verification, the UE performs sequential steps to verify the alert source authenticity, freshness, and target alerting area. First, it compares the alert expiration time T_{exp} with its internal clock such that $T_{\text{current}} < T_{\text{exp}}$. This verification ensures the freshness of alerts. An adversary cannot broadcast an expired alert to the UE. Second, the UE validates the Target Area ID (A_{ID}) against its current location using GPS coordinates to prevent a wormhole attack. Existing PWS standardization already has such verification using Warning Area Coordinates for each $\text{SIB}_{\text{PWS}}^k$. In contrast, our scheme performs such verification prior to receiving and processing SIB_{PWS} to improve efficiency by avoiding processing invalid alert messages. Finally, the UE verifies the digital signature σ using PK_{CBE} .

$$\text{Verify}(\sigma, \langle C, A_{\text{ID}}, T_{\text{exp}} \rangle, PK_{\text{CBE}}) \stackrel{?}{=} 1$$

If the verification result outputs 1, it indicates the alert source authenticity. Because only the legitimate CBE can generate σ , the alert originates from the legitimate CBE.

If the security header received in SIB1 is valid, then the UE accepts the root commitment, C , as a trust anchor for subsequent SIB_{PWS} transmission. For the first alert broadcast, $\text{SIB}_{\text{PWS}}^1$, containing m_1 and H_{n-1} , the UE validates if $h(m_1 || H_{n-1}) \stackrel{?}{=} C$. This verification ensures m_1 is an authentic alert message generated by CBE, and UE stores H_{n-1} for the next SIB verification. The UE verifies the subsequent alert message segments, m_k , by validating $h(m_k || H_{n-k}) \stackrel{?}{=} H_{n-k+1}$. H_{n-k+1} is the hash stored in the previous message.

VI. FORMAL ANALYSIS OF T-PWS

In this section, we provide a rigorous formal analysis of the T-PWS using the Tamarin Prover to validate that our

design satisfies operator-agnostic security requirements (**S1-S4**) while enabling universal availability requirements (**F1-F2**) as described in Section IV-B. We open-source the modeling and results in a public repository for reproducibility [35].

A. Tamarin Prover

Tamarin Prover is a popular tool for cryptographic security protocol verification. Unlike finite-state model checkers, Tamarin automates the analysis of security protocols in a symbolic model for an unbounded number of concurrent sessions. It has been successfully utilized for formal verification of complex real-world security protocols including web security protocol TLS 1.3 [36], cellular networking protocol 5G-AKA [32], and 5G handover [37]. The formal verification of these standards has frequently resulted in the identification and mitigation of critical vulnerabilities prior to deployment [15].

The Tamarin prover takes three inputs: a protocol model, adversary specifications, and desired protocol security properties. It specifies protocol behavior using multiset rewrite rules that define state transitions through the consumption and generation of facts. The tool performs security analysis under the standard Dolev-Yao adversary model [30], granting the attacker full control over the network channel but constrained only by cryptographic hardness assumptions (e.g., signatures cannot be forged without private keys). The modeled protocol security requirements are formalized as *lemmas* in first-order logic that quantify over action facts in the system's execution traces. Tamarin employs constraint solving to exhaustively search the state space, either providing a mathematical proof that the properties hold or generating a counterexample trace that demonstrates a specific attack vector.

B. Modeling the Protocol

We model the T-PWS protocol described in Section V using 26 multiset rewrite rule interactions among five entities: UE, BS, AMF/CBC, UDM, and CBE. The model specification covers the complete lifecycle of the UE-network interaction, beginning with subscriber provisioning, where an abstracted 5G-AKA handshake establishes a NAS security key (K_{NAS}). This key is used to provision the CBE's public key (PK_{CBE}) to the UE, establishing a persistent root of trust.

To capture the *operator-agnostic* requirement, the alert verification phase models the UE with this persistent trust anchor PK_{CBE} . This simulates a device that verifies alerts from an untrusted foreign network using only PK_{CBE} . To ensure model convergence in Tamarin Prover, we bound the hash chain depth to $n = 3$, which provides an inductive proof of correctness for any $n > 3$.

Finally, we model the Dolev-Yao adversary capabilities strictly in accordance with our threat model (Section IV-A). We grant the adversary complete control over the air interface between the UE and BS, as this represents physically exposed attack surface in PWS, whereas internal core network channels (e.g., AMF-UDM, CBC-CBE) are abstracted as secure links. This modeling choice reflects 3GPP operational standards where core network links are protected by robust wired

IPsec/TLS protocols and radio broadcasting is designed as unauthenticated channel.

C. Verification Results

We formally verify the T-PWS design against functional and security requirements, described in Section IV-B using 5 lemmas.

1) *Functional Correctness to Satisfy F1-F2*: We verify that the architecture supports operator-agnostic alert verification. The lemma `lss_non_3GPP_executability` proves the existence of a valid trace where a UE successfully verifies a multi-segment alert broadcast by a foreign network different from its HN. This formally validates that the operator-agnostic design satisfies the universal availability requirement (**F1**) and that alert verification is decoupled from network authentication (**F2**). Additionally, two supporting lemmas confirm the general executability of the protocol architecture, validating that the system functions correctly under normal operating conditions.

2) *Security Validation to Satisfy S1-S4*: We verify four core security lemmas to confirm that T-PWS satisfies the threat model described in Section IV-A.

Secure Trust Bootstrapping (S1): The lemma `cbe_key_integrity` proves that for any UE in the normal state, the installed PK_{CBE} is authentic. The solver confirms that because the provisioning message relies on the NAS key (K_{NAS}), derived from the shared credentials (R, K), a Dolev-Yao adversary cannot derive keys without breaking the underlying cryptography of 5G-AKA protocol.

Source Authentication (S2): The `alert_integrity` lemma validates injective agreement: a UE only accepts an alert M if it was generated by the CBE. The proof (verified in 46 steps) holds because the ECDSA signing key SK_{CBE} is modeled as non-extractable. Consequently, the adversary cannot forge a valid signature σ , and thus cannot generate a valid commitment C corresponding to a fabricated message M' , ensuring that every accepted alert originates from the legitimate authority.

Geo-Fencing (S3): The `wormhole_resistance` lemma provides the definitive test for contextual integrity via alert geo-fencing. It confirms that a UE rejects valid alerts if the location context (A_{ID}) does not match the UE's actual physical location. Since the Area ID is cryptographically bound within the signed commitment C , the proof demonstrates that an adversary cannot tamper with or spoof the location field, effectively preventing wormhole attacks.

Replay Resistance (S4): The `replay_resistance` lemma proves that UEs reject alerts with expired timestamps. This property is enforced by the consumption of linear time facts in the model. Once the system clock transitions to a new epoch, the tokens required to verify previous timestamps are permanently removed, mathematically preventing the acceptance of replayed alerts.

VII. IMPLEMENTATION AND RESULT EVALUATION

We prototype our proposed T-PWS in opensource 4G/5G software to analyze the performance of the protocol based on the requirements described in Section IV-B3.

TABLE I: Computation time (ms) overhead of cryptographic operations (mean $\pm$ SD)

CBE (Generation)		UE (Verification)	
Hash Chain	Signature	Signature	Hash (per SIB)
0.058 $\pm$ 0.009	0.559 $\pm$ 0.115	1.74 $\pm$ 0.106	0.058 $\pm$ 0.018

A. Implementation and Testbed Setup

1) *Software Stack for Protocol Implementation*: We implemented the protocol using srsRAN [38] for UE and base station, and Open5GS [39] for core network. The NAS layer handlers in both Open5GS and srsRAN were modified to validate PK_{CBE} delivery immediately following 5G-AKA. Due to limited 5G SIB support in the current srsRAN release, the alert broadcasting and verification primitives were implemented on the 4G-LTE stack. This abstraction is valid for performance analysis since the PWS scheduling mechanisms are functionally equivalent between LTE and 5G-NR.

2) *Hardware*: We simulate the network entities (CBE, AMF, BS) on a workstation with an AMD Ryzen 7 5700U (16 GB RAM). The UE functionality was simulated on a MiniPC with an Intel N95 processor (8 GB RAM). We use two USRP B210 software-defined radios (SDRs) connected to both hardware devices for the RF front end.

3) *Experimental Setup*: We evaluate the prototype by analyzing UE event logs and packet captures (PCAP) at the radio interface. To assess architectural robustness under varying radio channel conditions, we control the Signal-to-Noise Ratio (SNR) by adjusting the Base Station transmit gain over a fixed 5-meter Line-of-Sight (LOS) radio link. We include alert reception probability, cryptographic verification latency, and total end-to-end processing delay as key performance metrics for performance evaluation which are discussed in the following sections.

B. Computational Overhead Analysis

In this section, we evaluate the computational performance of T-PWS to ensure that it meets the strict latency (**P1**) and energy efficiency (**P3**) constraints of critical public warning systems. We characterize the overhead using micro-benchmarks of cryptographic primitives and an analytical model of end-to-end delivery. Our analysis demonstrates that by decoupling segment verification from expensive digital signatures, T-PWS minimizes the computational burden on resource-constrained UEs, ensuring scalable and rapid alert verification compared with baseline security approaches.

1) *Computational Overhead for Cryptographic Operations*: We measure the execution time of the cryptographic operations during the protocol run in the experimental testbed described in Section VII-A. Table I summarizes the computational overhead by presenting the mean execution time and standard deviation (SD) for each operation. The results highlight a significant asymmetry between the symmetric and asymmetric cryptographic costs. On the UE side, digital signature verification (≈ 1.74 ms) is approximately 30 times more computationally expensive than verifying a single hash (≈ 0.058 ms). Similarly,

TABLE II: Communication overhead for each SIB type

SIB Type	Parameter	Size (Bytes)	Description
SIB1	Signature (σ)	64	ECDSA-256
	Timestamp (T_{exp})	8	64-bit UTC
	Area ID (A_{ID})	8	GPS Geo-Fence
	Root Commitment (C')	32	SHA-256 Hash
	Total Overhead	112	Per SIB1 Cycle
SIBPWS	Hash Pre-image (H_i)	32	SHA-256 Hash

on the CBE side, hash chain computation is approximately 10 times more efficient than signature generation. This sharp contrast in computing overhead provides an empirical basis for our architectural decision to utilize reverse hash chains for segment verification rather than relying on expensive digital signatures for every broadcast message.

2) *End-to-End Latency Analysis*: We provide an end-to-end latency analysis using the data shown in Table I to evaluate the protocol performance in time-sensitive multi-segment alert delivery. We denote the cumulative cryptographic latency, $L(n)$, as the sum of the generation time, T_{gen} at the CBE and the verification time, T_{ver} , for n -segment alerts at the UE. For the baseline security scheme, both CBE and UE perform computation-heavy digital signature operations (T^{sig}) as follows:

$$L_{Base}(n) = n \cdot (T_{gen}^{sig} + T_{ver}^{sig}) \approx n \cdot 2.30 \text{ ms}$$

In contrast, the T-PWS protocol requires only one digital signature operation, followed by n lightweight hash computations. Latency can be expressed as follows:

$$\begin{aligned} L_{TPWS}(n) &= (T_{gen}^{sig} + T_{ver}^{sig}) + n \cdot (T_{gen}^{hash} + T_{ver}^{hash}) \\ &\approx 2.30 \text{ ms} + n \cdot 0.12 \text{ ms} \end{aligned}$$

For a typical multi-page alert with $n = 10$ segments, the baseline adds approximately 23.0 ms end-to-end latency, whereas T-PWS adds only approximately 3.5 ms. Therefore, T-PWS reduces the latency in cryptographic computation by 85%, satisfying the performance requirement **P1**.

3) *Complexity and Energy Implications*: To generalize computation overhead results, we analyze the complexity using Big O notation and derive the energy consumption implication at the UE. Let C_{ds} denote the digital signature operation cost and C_h denote the hash function operation cost. The computational complexity of the baseline security scheme is $\mathcal{O}(n \cdot C_{ds})$, whereas for the T-PWS, it is $\mathcal{O}(1 \cdot C_{ds} + n \cdot C_h)$. From Table I, we find $C_h \ll C_{ds}$, therefore T-PWS effectively reduces the complexity of the dominant cryptographic operations from linear $\mathcal{O}(n)$ to constant $\mathcal{O}(1)$. Because energy consumption is proportional to execution time, the T-PWS protocol consumes significantly less energy at the UE to verify alerts. The Big O notation, $\mathcal{O}(1)$, implies that the heavy computational cost remains constant regardless of the number of alert segments, thereby satisfying the energy efficiency requirement **P3**.

C. Communication Overhead Analysis

We analyze the communication overhead to ensure that the additional protocol metadata of the T-PWS fits within the SIB

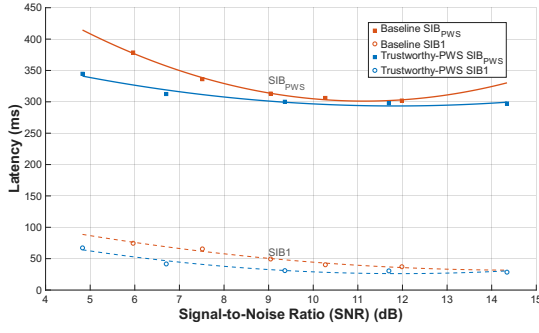


Fig. 5: Impact of radio channel quality (SNR) on SIB reception latency.

message constraints. Furthermore, we present an experimental analysis of the impact of communication overhead on alert reception latency at the UE.

1) *Communication Overhead*: Table II shows the detailed communication overhead for each SIB type in T-PWS. We employ the ECDSA-256 digital signature and SHA-256 hashing algorithms. Our protocol introduces the following additional fields: a digital signature (σ) of 64 bytes, an alert expiration timestamp (T_{exp}) of 8 bytes, a geo-fence area ID (A_{ID}) of 8 bytes, and a root commitment (C) of 8 bytes. The total overhead of 112 bytes fits within the 372 bytes payload limit of a standard SIB1 message. Additionally, for the alert payload segments, our protocol adds only 32 bytes for SHA-256 hash within each SIB_{PWS} , leaving ample space (300+ bytes) for actual alert content. For both SIB1 and SIB_{PWS} , our protocol supports the additional fields within the 3GPP physical layer constraints, satisfying performance requirement **P2**.

2) *Wireless Experimental Analysis*: Using the wireless experimental setup described in Section VII-A, We experimentally evaluate the impact of this overhead on alert reception latency by comparing T-PWS with the current 3GPP standard PWS (Baseline). We measure the reception time for SIB1 and the final segment (out of three) of a SIB_{PWS} at the UE under varying radio channel conditions. To control the channel quality, we adjusted the Base Station transmit power from 63 dB to 75 dB in 3 dB increments and recorded the resulting Signal-to-Noise Ratio (SNR).

Figure 5 shows the experimental results. Owing to the stochastic nature of the wireless channel, slight SNR variations occurred between the samples. We apply curve fitting to normalize the comparison. The results show that the SIB reception curves for the T-PWS and Baseline converge at a high SNR, indicating that the additional byte overhead does not introduce a statistically significant latency in alert reception. This confirms that the protocol maintains the real-time performance required for emergency warnings, satisfying requirement **P1**.

D. Geographical Coverage of T-PWS vs. Operator-Based PWS Authentication for Universal Availability

T-PWS enables PWS authentication delivered by *any* operators and enhances universal availability, i.e., a UE can receive

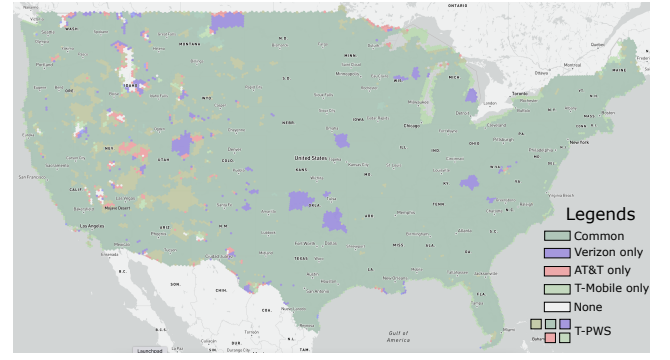


Fig. 6: Coverage map of T-PWS vs. three major mobile network operators in the US to illustrate universal availability. The original data and map is based on [40]. T-PWS covers all color region in the map except white (no network coverage by any major operators).

a PWS as long as *any* base station is within its reach. To support such objective, T-PWS uses the CBE's public key to authenticate the PWS during alert (Section V), in contrast to the previous research using an operator's public key (Section III-C). CBE initiating and authorizing the PWS uses all of the major operators to deliver the PWS alerts. To highlight such design impact and compare with the existing research, we simulate the geographical coverage of T-PWS vs. the previous research approach based on the subscribed operator acting as the trust source. Using FCC broadband map [40], in USA, Figure 6 shows the geographical coverage of the PWS alert using the three major operators (Verizon, AT&T, T-Mobile) vs. T-PWS (using 52 operators active across the nation as of June 30, 2025 [40]). Verizon has a geographical coverage of 60.02% across the nation; AT&T has 57.23%; T-Mobile has 45.02%; and so on. Previous research leveraging the CN of a singular operator (to which the UE is subscribed) has the corresponding geographical coverage. T-PWS using all the major operators has a significant wider geographical coverage of 69.38% across the nation, providing a more robust and wider coverage for universal availability.

VIII. CONCLUSION

In this research, we introduce Trustworthy-PWS, an operator-agnostic resilient PWS security architecture designed to resolve the fundamental PWS Trilemma in 5G networks. By decoupling the trust anchor from the mobile network operator and establishing the Cell Broadcast Entity (CBE) as the independent root of trust, our solution ensures that emergency alerts are universally verifiable by all users, including roaming, non-3GPP, and limited-service state devices, thereby maintaining the critical requirement of universal availability. We developed a lightweight commitment-based protocol that utilizes reverse hash chains to achieve end-to-end source authentication and message integrity. Formal verification using the Tamarin Prover confirmed that our proposed architecture successfully defends against Dolev-Yao adversaries, including sophisticated threats from compromised base stations and

wormhole attacks. Furthermore, our prototype implementation on a software-defined radio testbed demonstrated that the protocol reduced cryptographic processing latency by 85% compared to standard signature-based approaches, while the computational complexity reduced from $\mathcal{O}(n)$ to $\mathcal{O}(1)$. These results validate that Trustworthy-PWS is a practical, standard-compliant solution that secures public warning systems without compromising the speed or reach of life-saving alerts.

IX. ACKNOWLEDGMENT

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REFERENCES

- [1] 3GPP. TS 23.041 version 18.2.0, “Technical realization of Cell Broadcast Service (CBS),” 2023.
- [2] 3GPP. TS 22.268 version 18.3.0, “Public Warning System (PWS) requirements,” 2023.
- [3] FEMA, “Wireless Emergency Alerts,” 2023. [Online]. Available: <https://www.fema.gov/emergency-managers/practitioners/integrated-public-alert-warning-system/public/wireless-emergency-alerts>
- [4] 3GPP. TS 33.501 version 16.13.0, “Security architecture and procedures for 5G System,” 2023.
- [5] 3GPP. TS 33.331 version 17.2.0, “5G NR; Radio Resource Control (RRC); Protocol specification,” 2022.
- [6] E. Bitsikas and C. Pöpper, “You have been warned: Abusing 5g’s warning and emergency systems,” in *Proceedings of the 38th Annual Computer Security Applications Conference*, 2022, pp. 561–575.
- [7] G. Lee, J. Lee, J. Lee, Y. Im, M. Hollingsworth, E. Wustrow, D. Grunwald, and S. Ha, “This is your president speaking: Spoofing alerts in 4g lte networks,” in *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services*, 2019.
- [8] S. R. Hussain, M. Echeverria, A. Singla, O. Chowdhury, and E. Bertino, “Insecure connection bootstrapping in cellular networks: the root of all evil,” in *Proceedings of the 12th conference on security and privacy in wireless and mobile networks*, 2019, pp. 1–11.
- [9] A. J. Ross, B. Reaves, Y. Nasser, G. Cukierman, and R. P. Jover, “Fixing insecure cellular system information broadcasts for good,” in *Proceedings of the 27th International Symposium on Research in Attacks, Intrusions and Defenses*, 2024, pp. 693–708.
- [10] S. Purification, S. Wuthier, J. Kim, I. Kim, and S.-Y. Chang, “Base station certificate and authentication for 5g radio control security,” in *2025 IEEE 22nd International Conference on Mobile Ad-Hoc and Smart Systems (MASS)*, 2025.
- [11] 3GPP. TR 33.809 version 0.8.0, “Study on 5G security enhancements against false base stations,” online, 2021.
- [12] A. Perrig, R. Canetti, D. Song, and J. D. Tygar, “Efficient and secure source authentication for multicast,” in *Network and distributed system security symposium, NDSS*, vol. 1, no. 2001, 2001, pp. 35–46.
- [13] A. Perrig, J. Tygar, A. Perrig, and J. Tygar, “Tesla broadcast authentication,” *Secure Broadcast Communication: In Wired and Wireless Networks*, pp. 29–53, 2003.
- [14] Cybersecurity and Infrastructure Security Agency, “5G Potential Threat Vectors,” U.S. Department of Homeland Security, Technical Report, May 2021. [Online]. Available: <https://www.cisa.gov/resources-tools/resources/5g-potential-threat-vectors>
- [15] D. Basin, C. Cremers, J. Dreier, and R. Sasse, “Tamarin: verification of large-scale, real-world, cryptographic protocols,” *IEEE Security & Privacy*, vol. 20, no. 3, pp. 24–32, 2022.
- [16] S. R. Hussain, M. Echeverria, O. Chowdhury, N. Li, and E. Bertino, “Privacy attacks to the 4g and 5g cellular paging protocols using side channel information,” *Network and distributed systems security (NDSS) symposium2019*, 2019.
- [17] S. Hussain, O. Chowdhury, S. Mehnaz, and E. Bertino, “Lteinspector: A systematic approach for adversarial testing of 4g lte,” in *Network and Distributed Systems Security (NDSS) Symposium 2018*, 2018.
- [18] S. R. Hussain, M. Echeverria, I. Karim, O. Chowdhury, and E. Bertino, “Sgreasoner: A property-directed security and privacy analysis framework for 5g cellular network protocol,” in *Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security*, 2019.
- [19] E. Bitsikas and C. Pöpper, “Don’t hand it over: Vulnerabilities in the handover procedure of cellular telecommunications,” in *Annual Computer Security Applications Conference*, 2021, pp. 900–915.
- [20] A. Shaik, R. Borgaonkar, N. Asokan, V. Niemi, and J.-P. Seifert, “Practical Attacks Against Privacy and Availability in 4G/LTE Mobile Communication Systems,” *Network and Distributed System Security Symposium (NDSS)*, 5 2017.
- [21] A. Shaik, R. Borgaonkar, S. Park, and J.-P. Seifert, “New vulnerabilities in 4g and 5g cellular access network protocols: exposing device capabilities,” in *Proceedings of the 12th Conference on Security and Privacy in Wireless and Mobile Networks*, 2019, pp. 221–231.
- [22] European Parliament and Council of the European Union, “Directive (EU) 2018/1972 of the European Parliament and of the Council,” <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32018L1972>, 2018, [Article 110].
- [23] Federal Communications Commission, “Wireless Emergency Alerts; Amendments to Part 11 of the Commission’s Rules Regarding the Emergency Alert System (Report and Order),” FCC 16-127, PS Docket Nos. 15-91, 15-94, Nov. 2016, see paragraphs 50–53 on Geo-Targeting latency constraints. [Online]. Available: <https://docs.fcc.gov/public/attachments/FCC-16-127A1.pdf>
- [24] A. Singla, R. Behnia, S. R. Hussain, A. Yavuz, and E. Bertino, “Look before you leap: Secure connection bootstrapping for 5g networks to defend against fake base-stations,” in *Proceedings of the 2021 ACM Asia Conference on Computer and Communications Security*, 2021.
- [25] S. Purification and S.-Y. Chang, “Verifiable alerts for 4g/5g public warning system,” in *2025 IEEE Conference on Communications and Network Security (CNS)*, 2025, pp. 1–9.
- [26] A. Perrig, R. Canetti, J. D. Tygar, and D. Song, “Efficient authentication and signing of multicast streams over lossy channels,” in *Proceeding 2000 IEEE symposium on security and privacy. S&P 2000*. IEEE, 2000, pp. 56–73.
- [27] N. Haller, “The s/key one-time password system,” Tech. Rep., 1995.
- [28] L. Lamport, “Password authentication with insecure communication,” *Commun. ACM*, vol. 24, no. 11, p. 770–772, Nov. 1981. [Online]. Available: <https://doi.org/10.1145/358790.358797>
- [29] 3GPP. TR 33.969 version 18.0.0 Release 18, “Study on security aspects of Public Warning System (PWS),” online, 2024.
- [30] D. Dolev and A. Yao, “On the security of public key protocols,” *IEEE Transactions on information theory*, vol. 29, no. 2, pp. 198–208, 2003.
- [31] S. Kent and K. Seo, “Security Architecture for the Internet Protocol,” RFC 4301, Dec. 2005. [Online]. Available: <https://www.rfc-editor.org/info/rfc4301>
- [32] D. Basin, J. Dreier, L. Hirschi, S. Radomirovic, R. Sasse, and V. Stettler, “A formal analysis of 5g authentication,” in *Proceedings of the 2018 ACM SIGSAC conference on computer and communications security*, 2018, pp. 1383–1396.
- [33] FEMA, “Integrated Public Alert & Warning System (IPAWS),” <https://www.fema.gov/emergency-managers/practitioners/integrated-public-alert-warning-syste>, [].
- [34] V. Dukhovni, “Opportunistic security: Some protection most of the time,” RFC 7435, December 2014. [Online]. Available: <https://datatracker.ietf.org/doc/html/rfc7435>
- [35] Sourav Purification, “Tamarin Model for Trustworthy-PWS,” <https://github.com/NSSL-UCCS/Trustworthy-PWS>, 2026.
- [36] C. Cremers, M. Horvat, J. Hoyland, S. Scott, and T. Van Der Merwe, “A comprehensive symbolic analysis of tls 1.3,” in *Proceedings of the 2017 ACM SIGSAC conference on computer and communications security*, 2017, pp. 1773–1788.
- [37] A. Peltonen, R. Sasse, and D. Basin, “A comprehensive formal analysis of 5g handover,” in *Proceedings of the 14th ACM conference on security and privacy in wireless and mobile networks*, 2021, pp. 1–12.
- [38] Software Radio Systems, “srsRAN 4G,” https://github.com/srsran/srsRAN_4G/releases/tag/release_22_10, 2023.
- [39] Sukchan Lee, “Open5GS,” <https://github.com/open5gs>, 2022.
- [40] Federal Communications Commission, “FCC Broadband Map,” <https://broadbandmap.fcc.gov/home>, 2025, [].